Supplement to: Classical and Self-Supervised Hankel Filtering for Magnitude-Only Rydberg Channel Estimation

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***Abstract*—This supplement records the completed estimator checks, broad OptShrink comparisons and retained negative experiments. Every study keeps its own data and evaluation rule. The original trained models are unchanged. Fresh test banks strengthen the comparison, but do not measure uncertainty from training on new datasets. Exploratory results do not establish convergence, hardware performance or a general reliability guarantee.**

I. Broad OptShrink Comparisons: Full Numerical Record

Study A tunes OptShrink on the original 2,048 validation observations and masks. Study B inherits that lookup and selects rank from each new block of observations. Neither study retrains the compact estimator. Each has two independent test banks, 126 primary cells and two separate stress cells. All 30 pilots are reused for the final fit, without additional observations. The inference lookup uses supplied noise and reference strength, not true channel power or path count.

Study A uses test-bank seeds 202610210001 and 202610310001; Study B uses 202610610001 and 202610710001. Each cell adds 100 times its zero-based index to the bank seed. Both use 16 worlds per cell and bank. The selected-model seeds are 9101, 9102 and 9103. Sources, settings and checkpoints are hashed before test generation. Settings remain frozen after the test data are inspected.

The primary score averages 126 per-cell dB ratio-of-sums NMSE values, after pooling the two equal-sized test banks within each cell. It is not the dB of one pooled error ratio. Learned scores average the three separate model scores. Bootstrap resampling preserves pairing within each bank/cell; confidence intervals condition on these weights and the offline validation choices. They do not cover repeated training or offline selection uncertainty.

A. Declared Contrasts and All Comparators

The numerical tables separate the declared average-dB contrasts, alternative error summaries, individual models, test banks and stress cases. These summaries are not interchangeable: a gain in one average does not establish a gain in every cell or at every error threshold. All individual-cell scores are retained in the evidence records.

B. Frozen Settings and Numerical Verification

The online selector searches a shared rank, not separate ranks for each user. Its mixing strength and iteration count are inherited from the matching in-loop or final-only row in the offline lookup. Thus the rank comparison does not exhaust all possible classical policies. Candidate initialization uses only the first 21 pilots. After scoring the other nine, the winner starts again from all 30. Joint selection evaluates both placements and counts both searches in timing. Search work shared while generating several experimental arms is not used to understate standalone timing.

Study A: all 4,096 test worlds regenerated exactly; 3,584 representative predictions replayed. Maximum recorded independent classical absolute output difference was 5.73×10^{-13}. An independent scalar-loop calculation reproduced every primary score. These checks test the saved implementation and data, not an external reproduction of the study.

Study B: all 4,096 test worlds regenerated exactly; 2,816 representative predictions replayed. Maximum recorded independent rank-selector absolute output difference was 1.26×10^{-12}. An independent scalar-loop calculation reproduced every primary score. These checks test the saved implementation and data, not an external reproduction of the study.

Study A also regenerated all 2,048 validation observations and replayed selection from saved scores. Study B independently checked both rank-score vectors, chosen settings and all-pilot refits with NumPy/SciPy. Its new audit did not independently replay HS-Rician again; that backend check is in Study A. Both studies passed finite-output, source-hash, checkpoint-hash and freeze-before-generation checks. No failed trial was removed. Full per-cell values and source records are supplied as JSON evidence in the manuscript archive; full observation and prediction arrays remain in the experimental archive.

II. Retained Negative Follow-Ups

These studies tested possible improvements before the broad comparisons. None changed the three original 188-parameter checkpoints. Their outcomes are kept separate because they use different test banks and budgets.

A. A Controller Around OptShrink

The first hybrid used a learned mixing rule around OptShrink. On 704 fresh worlds, its seven-SNR average was -14.129 dB, compared with -14.535 dB for the stronger OptShrink control. It therefore lost 0.406 dB. Its ten-stage schedule and some ranks did not match the strong baseline.

TABLE I
LEARNED GAIN (POSITIVE FAVORS LEARNING), THREE-CONTRAST-ADJUSTED 95% INTERVALS WITHIN EACH STUDY, AND LARGEST OBSERVED PRIMARY-CELL DETERIORATION. CELLWISE SAFETY IS NOT ESTABLISHED.

Study	Comparator	Gain (dB)	Adjusted interval	Worst harm (dB)
A	Offline OptShrink, selected placement	0.537	[0.450, 0.695]	3.117
A	Offline OptShrink, in-loop	0.565	[0.477, 0.723]	3.117
A	Offline OptShrink, final-only	0.738	[0.621, 0.925]	3.686
B	Online rank + placement selection	-0.449	[-0.569, -0.358]	4.478
B	Online rank, in-loop	-0.507	[-0.562, -0.437]	5.495
B	Online rank, final-only	-0.076	[-0.226, 0.025]	3.445

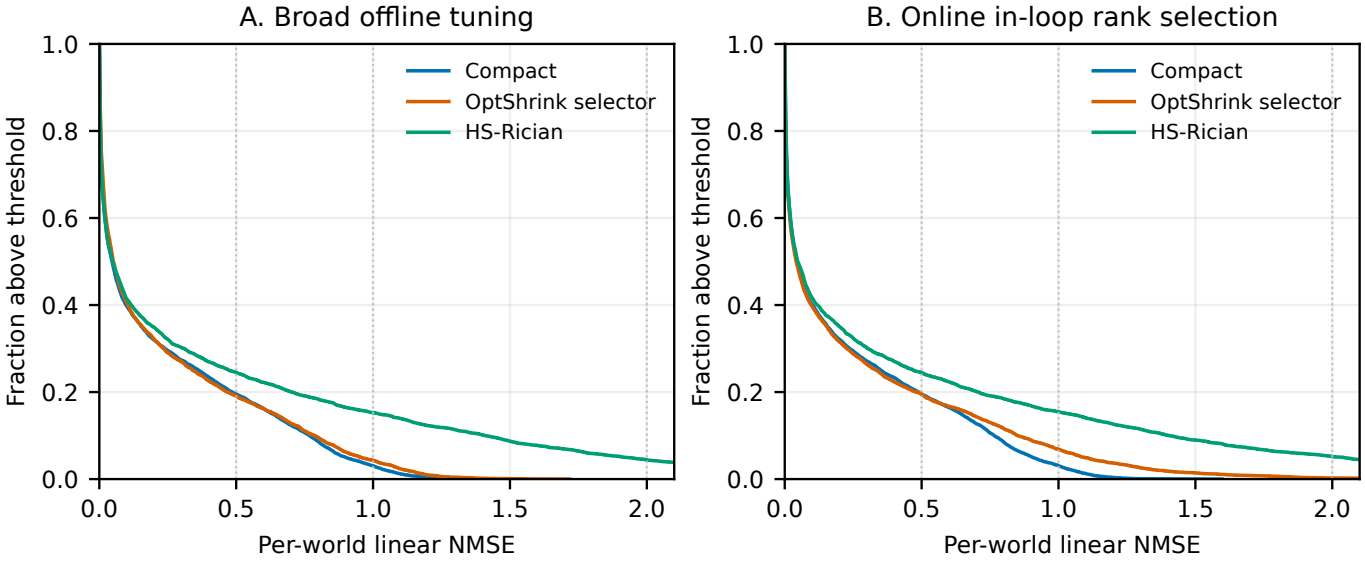


Fig. 1. Primary empirical exceedance curves over the displayed threshold range, separately by study. Learned curves average the three model-specific empirical distributions on the same worlds; predictions are not independent channels. All values, including larger errors, are retained in the evidence arrays. A lower rate at one threshold does not imply a lower rate at every threshold.

TABLE II
STUDY A: ALL PRIMARY AVERAGES AND MEAN END-TO-END CPU TIME. FIXED-BUDGET LABELS DENOTE EM UPDATES, NOT MATCHED RUNTIME.

Method	NMSE (dB)	Time (ms)
Offline OptShrink, selected placement	-13.028	15.61
Offline OptShrink, in-loop	-13.000	15.57
Offline OptShrink, final-only	-12.827	6.16
OptShrink, 5 updates	-9.816	2.47
OptShrink, 10 updates	-12.336	4.56
OptShrink, 20 updates	-12.926	8.73
OptShrink, 50 updates	-12.941	21.20
OptShrink, 100 updates	-12.951	41.88
Original narrow OptShrink lookup	-11.880	6.31
EM-GS, 100 updates	-9.922	13.08
HS-Rician, one sweep	-12.996	98.91
Compact estimator (188 weights)	-13.565	4.90

TABLE III
STUDY B: ALL PRIMARY AVERAGES AND MEAN END-TO-END CPU TIME. FIXED-BUDGET LABELS DENOTE EM UPDATES, NOT MATCHED RUNTIME.

Method	NMSE (dB)	Time (ms)
Online rank, in-loop	-14.154	108.70
Online rank, final-only	-13.723	13.96
Online rank + placement selection	-14.096	114.17
Offline OptShrink, selected placement	-13.196	15.76
Offline OptShrink, in-loop	-13.151	15.68
Offline OptShrink, final-only	-12.863	6.23
Original narrow OptShrink lookup	-12.041	6.38
EM-GS, 100 updates	-9.958	13.15
HS-Rician, one sweep	-13.059	99.88
Compact estimator (188 weights)	-13.647	4.96

A clipped gate also had zero local derivative in many high-SNR cases. These were reasons to repair the comparison, not evidence that repairing it would produce an improvement.

The corrected pilot used the strong baseline's ranks and noise-dependent iteration schedule, removed the hard clipping, and trained two initializations on one shared dataset. Settings were fixed before 704 new test worlds were generated. The learned gate started at 0.95 where the baseline used strength one; a near-initial control accounted for this difference. Table X gives the primary seven-SNR averages. Four stress cells were reported separately.

The corrected hybrid gained only 0.019 dB over the strong starting baseline, with conditional pointwise 95% interval $[0.004, 0.033]$ dB. It lost 0.022 dB to retuned classical mixing and 0.040 dB to the simpler stage-only learned rule. Mean same-backend processing times were about 7.2 ms for the hybrid and 6.2 ms for OptShrink. The corrected gate had no exact-zero local slopes in the recorded tests, but this did not make it the better estimator. The screen failed, and the

TABLE IV
ALTERNATIVE SUMMARIES ON PRIMARY WORLDS. POOLED NMSE IS IN DB; PER-CASE MEAN NMSE IS LINEAR. LAST THREE COLUMNS ARE PERCENTAGES ABOVE THE FIXED THRESHOLDS. ZERO OBSERVED EXCEEDANCES DO NOT IMPLY ZERO POPULATION RISK.

Study	Method	Pooled	Case mean	> 0.5	> 1	> 2
A	Compact estimator (188 weights)	-6.704	0.221	19.54	3.03	0.00
A	Offline OptShrink, selected placement	-6.588	0.227	19.12	4.34	0.00
A	HS-Rician, one sweep	-4.216	0.399	24.48	15.23	4.37
A	EM-GS, 100 updates	-1.415	0.777	30.38	23.59	14.71
B	Compact estimator (188 weights)	-6.683	0.221	19.55	3.08	0.00
B	Online rank, in-loop	-6.217	0.247	19.52	6.87	0.22
B	HS-Rician, one sweep	-4.166	0.402	24.50	15.48	5.21
B	EM-GS, 100 updates	-1.392	0.778	30.63	23.26	14.96

TABLE V
TRAINING-SEED AND TEST-BANK SUMMARIES, KEPT DISTINCT. BANK DB SCORES NEED NOT AVERAGE TO THE SCORE OBTAINED AFTER POOLING THEIR LINEAR ERRORS.

Study	Subset	NMSE (dB)
A	Model 9101	-13.676
A	Model 9102	-13.327
A	Model 9103	-13.692
A	Compact estimator (188 weights), bank 1	-13.700
A	Compact estimator (188 weights), bank 2	-13.559
A	Offline OptShrink, selected placement, bank 1	-13.059
A	Offline OptShrink, selected placement, bank 2	-13.064
B	Model 9101	-13.759
B	Model 9102	-13.417
B	Model 9103	-13.765
B	Compact estimator (188 weights), bank 1	-13.666
B	Compact estimator (188 weights), bank 2	-13.648
B	Offline OptShrink, selected placement, bank 1	-13.207
B	Offline OptShrink, selected placement, bank 2	-13.183
B	Online rank + placement selection, bank 1	-14.196
B	Online rank + placement selection, bank 2	-14.050
B	Online rank, in-loop, bank 1	-14.166
B	Online rank, in-loop, bank 2	-14.149

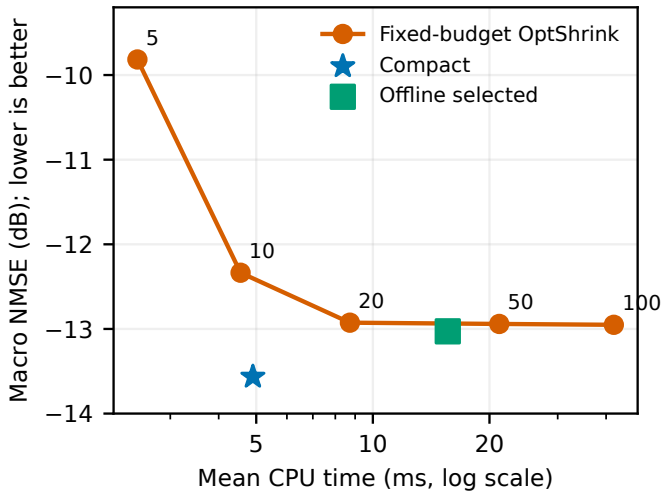


Fig. 2. Study A accuracy–cost curve. Labels are fixed update budgets, each tuned separately for placement, rank and mixing. The selected lookup may use different budgets by noise/reference pair. No equal-runtime optimum is claimed.

TABLE VI
STUDY A: DESCRIPTIVE MARGINAL AVERAGES OVER THE OTHER PRIMARY CONDITIONS. OPTSHRINK IS OFFLINE-SELECTED. THESE ARE NOT ADDITIONAL INDEPENDENT TESTS.

Study	Condition	Compact	OptShrink	Gain
A	SNR -10	-1.057	-0.835	0.222
A	SNR -5	-3.682	-3.894	-0.212
A	SNR 0	-8.346	-8.220	0.126
A	SNR 5	-13.481	-12.832	0.649
A	SNR 10	-18.636	-17.375	1.262
A	SNR 15	-23.521	-22.275	1.246
A	SNR 20	-26.232	-25.767	0.465
A	RSR 5	-12.494	-11.840	0.654
A	RSR 12	-13.968	-13.450	0.518
A	RSR 20	-14.234	-13.795	0.439
A	Paths 3–7	-14.295	-13.536	0.759
A	Paths 2	-15.863	-13.674	2.190
A	Paths 4	-14.704	-13.814	0.890
A	Paths 8	-13.099	-13.255	-0.156
A	Paths 12	-12.119	-12.309	-0.190
A	Paths 16	-11.310	-11.582	-0.272

hybrid was not expanded. Its numerical audit regenerated 2,720 worlds and checked 198 checkpoint predictions and 11 independent classical predictions.

B. Stopping, Filter Release and a Larger Learned Control

A separate 704-world pilot included two independent 352-world banks, each with 32 worlds in seven primary and four stress cells. The first bank tested observable stopping and releasing the structural filter. Frozen OptShrink, observed stopping and observed release reached -14.530, -14.498

TABLE VII
STUDY B: DESCRIPTIVE MARGINAL AVERAGES OVER THE OTHER PRIMARY CONDITIONS. OPTSHRINK IS ONLINE IN-LOOP RANK SELECTION. THESE ARE NOT ADDITIONAL INDEPENDENT TESTS.

Study	Condition	Compact	OptShrink	Gain
B	SNR -10	-1.073	-0.362	0.710
B	SNR -5	-3.605	-3.574	0.031
B	SNR 0	-8.284	-8.426	-0.142
B	SNR 5	-13.581	-13.888	-0.307
B	SNR 10	-18.694	-19.289	-0.595
B	SNR 15	-23.365	-24.210	-0.845
B	SNR 20	-26.926	-29.329	-2.403
B	RSR 5	-12.754	-13.098	-0.345
B	RSR 12	-13.985	-14.530	-0.544
B	RSR 20	-14.202	-14.834	-0.632
B	Paths 3–7	-14.342	-14.546	-0.203
B	Paths 2	-16.436	-17.151	-0.715
B	Paths 4	-14.653	-15.150	-0.498
B	Paths 8	-13.014	-13.495	-0.482
B	Paths 12	-12.210	-12.592	-0.382
B	Paths 16	-11.227	-11.990	-0.763

and -14.449 dB, respectively. Release lost 0.050 dB to observed stopping and 0.081 dB to frozen OptShrink. Stopping improved the weak-reference diagnostic, but not the primary average. It computed all 100 updates before choosing an output; its selected iteration count was not its total processing cost.

The second bank compared a newly trained compact model with an independently implemented exact-EM Transformer with 1,585,920 parameters. Both used 4,096 training cases, 12 epochs, matching masks and order, one training seed, and held-out likelihood for selection. Selected NMSEs were -14.325 dB for the compact model, -9.726 dB for the Transformer, -14.289 dB for OptShrink and -10.116 dB for EM100. The compact–OptShrink gain interval, $[-0.095, 0.176]$ dB, included zero. The Transformer was worse than EM100, so this pilot did not provide a strong learned comparator. It was not a reproduction of published URformer and supports no superiority claim over that method. Neither training budget establishes convergence.

C. An Error-Model Feasibility Check

A further screen examined whether local sensitivity could predict the error entering a Hankel filter. It used 24 fixed channel/pilot/reference worlds, 256 repeated noise draws and eight separate prediction anchors per world. Noise repetitions were not new channel realizations. The ordinary-start covariance check passed six of ten primary groups, below its required eight; the total-error/bias check passed seven. At SNR -10 dB after 100 EM updates, predicted variance was 16.48 times observed variance. With a weak reference it was 5.72 times. Estimated bias accounted for about 57% of total error at 20 updates and 22% at 100 updates in the weak-reference group.

Two supplementary finite-difference checks exceeded their original tolerance. Smaller steps and independent automatic differentiation resolved the numerical discrepancy. Ordinary-start relative Jacobian errors in the two problematic settings were about 0.0113% and 0.0023%, too small to explain the main prediction failure. The scientific screen remained failed. This study concerned unfiltered, row-separable EM; it did not validate an error model for interleaved Hankel reconstruction or establish an optimal shrinker. No new filter was promoted on the basis of these predictions.

D. What These Checks Leave Open

An observable likelihood score is not a channel-error guarantee. A smooth gate is not enough to establish useful adaptation. More network parameters do not establish a stronger trained baseline. The retained failures narrow the claim to the tested original compact estimator; they do not supply evidence for a new hybrid, an uncertainty-aware filter or a general improvement to OptShrink.

E. Retained Figures From the Earlier Studies

These plots retain the earlier narrow-population and operating-limit results. They use the protocols stated in their captions, not the new broad-study test banks.

III. MATCHED ONE-SWEEP BASELINE AND TAIL-ERROR AUDIT

This post-review follow-up uses the already inspected 1,152-world OptShrink bank: 128 worlds in each of seven primary SNR cells and two stress cells. It generates no new channels and trains no new model. The protocol is frozen before running the added comparator, not before the historical bank was inspected. It must not be described as a new blinded confirmation. The seven primary cells contain 896 distinct worlds.

The one-sweep HS-Rician estimator fits every rank 1–16 on the first 21 pilots for 25 EM updates, scores Rician likelihood on the other nine, then refits the selected rank with 100 updates and all 30 pilots. Each candidate and refit uses one Cadzow sweep. The sweep count is the only changed setting relative to the saved HS-Rician baseline; the new one-sweep candidates select their own ranks. Rank 16 bypasses filtering. No true channel is an estimator input. Learned checkpoints, OptShrink settings, initialization and observation budgets remain unchanged.

The added one-sweep HS-Rician control reaches -13.314 dB, improving on four sweeps by 0.424 dB. Learning retains a 1.004 dB average gain, with paired interval $[0.922, 1.088]$ dB. This closes the one-versus-four-sweep gap on the same bank, not the comparison to every possible hard-rank method. In the separate weak-reference and two-path cells, one sweep instead beats learning by 1.976 and 1.144 dB.

At the original threshold $e_i/d_i > 1$, the seven-SNR reduction in failure rate versus in-loop OptShrink is 4.13 percentage points, with paired interval $[3.051, 5.283]$. At -10 dB alone the interval is $[21.354, 36.979]$ percentage points. Learned seed-specific rates there are 5.47%, 5.47% and 7.81%, versus 35.16% for OptShrink.

The conclusion depends on the error threshold (Table XII). At threshold 0.5, OptShrink has fewer exceedances; at threshold 2, neither method has any observed exceedance. Zero

TABLE VIII
STRESS CONDITIONS, EXCLUDED FROM PRIMARY AVERAGES. WEAK REFERENCE: SNR 5/RSR 0 DB. NEARLY NOISELESS: SNR 60/RSR 12 DB. BOTH USE RANDOM THREE-TO-SEVEN-PATH CHANNELS.

Study	Condition	Method	NMSE (dB)
A	Weak reference	Compact estimator (188 weights)	-7.842
A	Weak reference	Offline OptShrink, selected placement	-11.905
A	Weak reference	EM-GS, 100 updates	-6.109
A	Nearly noiseless	Compact estimator (188 weights)	-6.693
A	Nearly noiseless	Offline OptShrink, selected placement	-66.886
A	Nearly noiseless	EM-GS, 100 updates	-65.604
B	Weak reference	Compact estimator (188 weights)	-7.787
B	Weak reference	Offline OptShrink, selected placement	-11.498
B	Weak reference	EM-GS, 100 updates	-6.261
B	Weak reference	Online rank + placement selection	-10.101
B	Weak reference	Online rank, in-loop	-10.101
B	Nearly noiseless	Compact estimator (188 weights)	-6.835
B	Nearly noiseless	Offline OptShrink, selected placement	-67.618
B	Nearly noiseless	EM-GS, 100 updates	-66.212
B	Nearly noiseless	Online rank + placement selection	-70.134
B	Nearly noiseless	Online rank, in-loop	-70.121

TABLE IX
BROAD OFFLINE SETTINGS. WITHIN EACH PLACEMENT, COLUMNS GIVE RANK, MIXING FACTOR AND UPDATE COUNT. THE SELECTED COLUMN CHOOSES BETWEEN THOSE PLACEMENTS. RANK ZERO DENOTES NO FILTERING.

SNR / RSR	Loop r	β	T	Final r	β	T	Selected
-10 / 5	1	0.75	5	2	1.0	50	final
-10 / 12	1	0.5	5	1	0.75	5	final
-10 / 20	1	0.75	5	1	0.75	5	loop
-5 / 5	2	0.5	10	3	0.75	20	final
-5 / 12	2	0.5	5	2	0.75	5	loop
-5 / 20	3	0.75	5	3	0.75	5	loop
0 / 5	4	0.5	20	5	0.75	100	loop
0 / 12	4	0.5	10	5	0.75	10	loop
0 / 20	4	0.5	10	7	1.0	5	loop
5 / 5	7	0.5	100	8	1.0	100	loop
5 / 12	7	0.5	20	8	1.0	20	loop
5 / 20	8	0.75	10	8	1.0	10	loop
10 / 5	9	1.0	100	9	0.75	100	loop
10 / 12	10	0.75	20	9	1.0	50	loop
10 / 20	10	1.0	20	10	1.0	20	loop
15 / 5	9	0.5	50	10	1.0	100	loop
15 / 12	10	0.75	50	10	1.0	50	loop
15 / 20	10	0.75	50	10	1.0	50	loop
20 / 5	11	1.0	100	13	1.0	100	loop
20 / 12	12	1.0	50	12	1.0	50	loop
20 / 20	11	0.75	50	11	1.0	50	loop

TABLE X
CORRECTED HYBRID PILOT: LOWER NMSE IS BETTER.

Method	NMSE (dB)
Strong OptShrink	-14.596
Retuned classical mixing	-14.637
Residual-based classical rule	-14.633
Stage-only learned mixing	-14.656
Corrected measurement-conditioned hybrid	-14.616

observed events do not establish zero population risk. These exploratory, pointwise intervals retain paired worlds and fixed models and are not adjusted for multiple thresholds. Learning therefore does not dominate the whole error distribution. The original severe-error definition and primary NMSE metric are unchanged.

The post-review timing run measures all five methods together on the same independent 24-world SNR 5 dB timing bank, with three repetitions. Mean times are 5.16 ms for one adaptive model, averaged across three trained models, 5.68 ms for in-loop OptShrink, 3.87 ms for final-only OptShrink, and 119.08/358.07 ms for one-/four-sweep HS-Rician. Initialization and setting selection or lookup are included; model loading and file I/O are excluded. These are new paired timing measurements, not averages with the earlier timing run. Backend differences remain. Learning avoids the hard method's per-case search, but has no large speed advantage over offline-tuned OptShrink.

A. What the Uncertainty Includes

The new audit uses 4,000 bootstrap draws, resampling 128 whole worlds within each cell. A draw keeps all paired methods and learned seed outputs together. NMSE first sums errors and channel energies within each cell, converts their

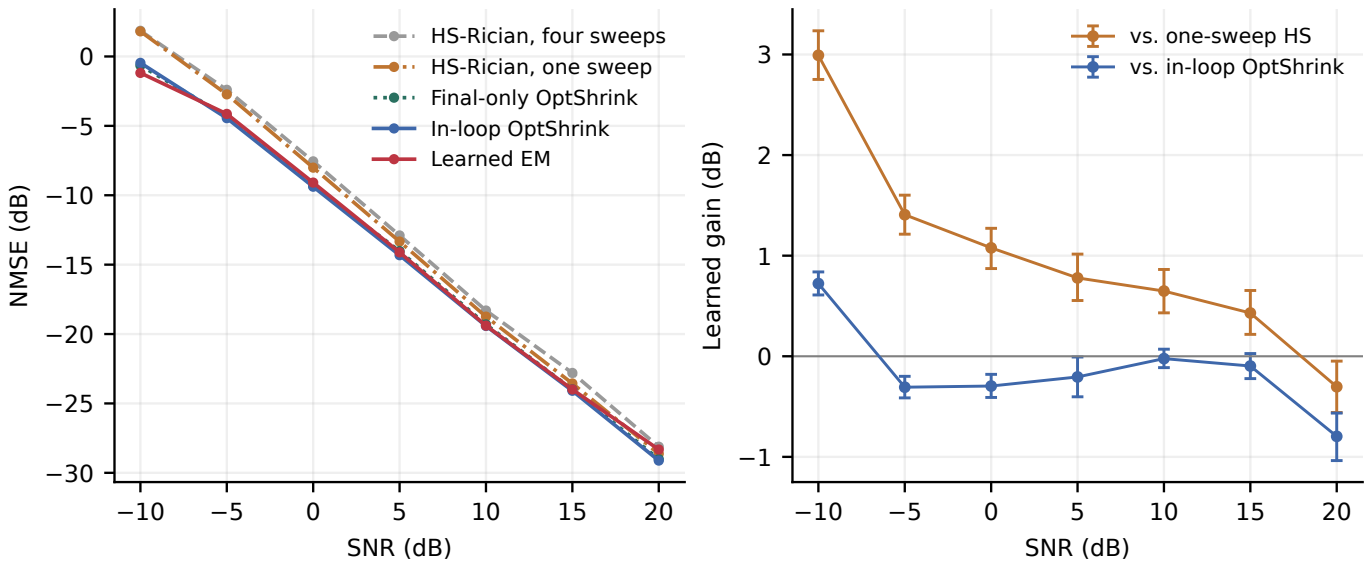


Fig. 3. Earlier narrow-population comparison: 128 paired trials per SNR and frozen learned models. The one-sweep HS-Rician result was added on the same saved worlds. Right: learned gains over one-sweep HS-Rician and in-loop OptShrink; positive favors learning. Intervals are pointwise and condition on the saved models.

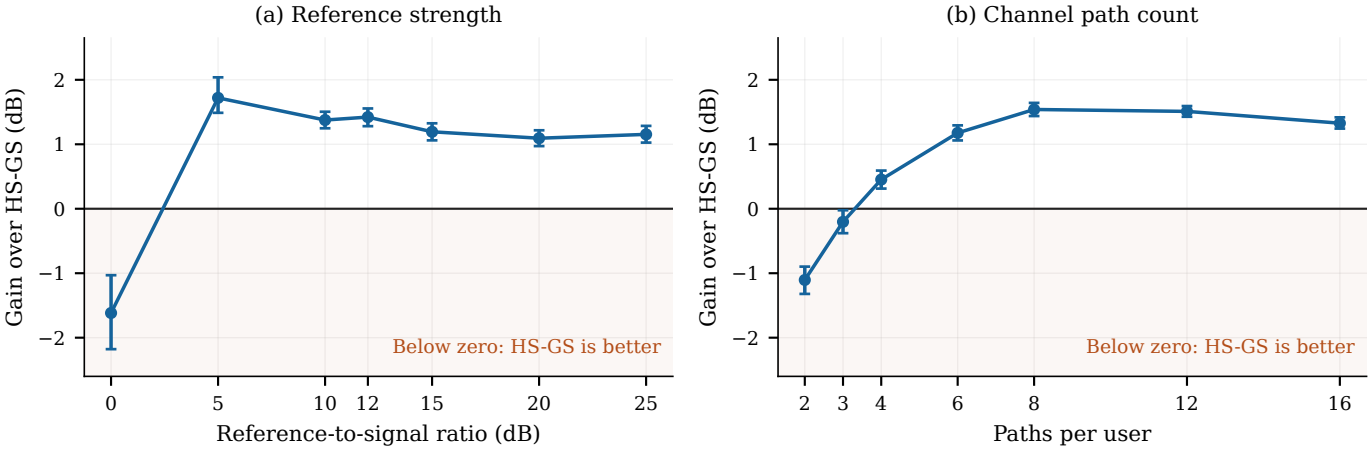


Fig. 4. Limits in the original learned study at SNR 5 dB. (a) Reference strength for three-to-seven-path channels. (b) Fixed path counts at RSR 12 dB. Gains compare adaptive EM with four-sweep HS-amplitude (HS-GS in the plot). Bars are conditional pointwise paired intervals; tested path counts overlap training.

TABLE XI
MATCHED SAVED-BANK RESULTS, LOWER DB NMSE IS BETTER. STRESS CELLS ARE EXCLUDED FROM THE SEVEN-SNR MACRO-AVERAGE. LEARNED VALUES AVERAGE THREE FIXED-MODEL DB SCORES, NOT ENSEMBLE ESTIMATES.

Cell	HS, one sweep	HS, four sweeps	In-loop OptShrink	Final OptShrink	Learned
SNR -10	1.808	1.850	-0.461	-0.650	-1.184
SNR -5	-2.731	-2.401	-4.445	-4.316	-4.138
SNR +0	-8.010	-7.558	-9.384	-9.333	-9.088
SNR +5	-13.335	-12.896	-14.320	-14.018	-14.114
SNR +10	-18.746	-18.306	-19.417	-19.306	-19.394
SNR +15	-23.561	-22.803	-24.088	-23.985	-23.991
SNR +20	-28.625	-28.118	-29.118	-29.010	-28.322
RSR 0 dB	-9.045	-8.307	-5.026	-6.613	-7.069
Two paths	-17.958	-17.833	-16.406	-16.592	-16.814
Seven-SNR mean	-13.314	-12.890	-14.462	-14.374	-14.319

ratio to dB, averages fixed-model scores, and then equally averages the seven cells. Failure rates instead count per-world error/energy ratios above the stated threshold and average fixed-model rates. Training seeds are not independent test

TABLE XII
POST-HOC ERROR-THRESHOLD SENSITIVITY ON THE STRONGER BANK. RATES AVERAGE SEVEN SNR CELLS; POSITIVE DIFFERENCES FAVOR LEARNING.

NMSE threshold	Learned (%)	In-loop OptShrink (%)	OptShrink minus learned 95% interval (points)
0.5	17.00	15.85	$[-1.935, -0.409]$
1	0.89	5.02	$[3.051, 5.283]$
2	0.00	0.00	—

observations. The intervals are pointwise, conditional on the saved models and exploratory. Thresholds 0.5 and 2 were added for sensitivity analysis; the original threshold 1 was not selected to maximize a new contrast. No risk dominance or safety-critical guarantee is inferred. A degenerate empirical bootstrap when there are no events is not evidence of known zero event probability; such intervals are omitted from the table.

B. Verification and Numerical Scope

The run checks original source and checkpoint hashes, saved observations and truth hashes, and all 1,152 trial identities. It recomputes the old prediction errors from channel arrays, rather than copying headline statistics. Four-sweep replay agrees exactly with the existing solver; full-rank bypass is exactly equal for one and four sweeps. Each added prediction retains its channel, selected rank, all candidate scores, trial ID and processing time. A run lock prevents duplicate writers, and completed cases are reused on resume. Earlier result files are not overwritten. The accompanying evidence archive contains this follow-up's protocol, code, source hashes, predictions and aggregate outputs. The large original observation/checkpoint bank is retained in the separate Paper 2 reproduction archive. These checks validate the recorded computation, not external replication or hardware performance.

The rest of this supplement retains the earlier learned-study evidence. Statements about its original timing samples, baselines and data counts refer to those separately labelled protocols, not the new follow-up.

IV. LOCAL RECOVERY AND REFERENCE CALIBRATION

For one array element, let $g, s_p \in \mathbb{C}^K$, with $\mu_p = g^T s_p + b_p$ and noiseless amplitude $a_p = |\mu_p|$. Use real coordinates $x = [\operatorname{Re} g^T, \operatorname{Im} g^T]^T$. When every $\mu_p \neq 0$, differentiation gives

$$da_p = \operatorname{Re}\{\mu_p^* s_p^T dg/|\mu_p|\}, \tag{1}$$

$$A_p = [\operatorname{Re} c_p^T, -\operatorname{Im} c_p^T], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{2}$$

Here $A \in \mathbb{R}^{P\times 2K}$ is the real amplitude Jacobian.

Local statement. If $\operatorname{rank} A = 2K$, the noiseless amplitude map is one-to-one in a sufficiently small neighborhood of this g. To see this, choose $2K$ independent rows. Their square Jacobian is invertible, so the inverse-function theorem supplies a local inverse for that measurement subset. The remaining measurements cannot remove this local injectivity. This is not a global uniqueness theorem: distant channels may give the same measurements. A rank-deficient Jacobian also does not, by itself, rule out all forms of nonlinear local injectivity.

For small perturbations at a full-rank point, least-squares inversion has first-order sensitivity bounded by $1/\sigma_{\min}(A)$. Thus $P \geq 2K$ is necessary for regular full-rank local recovery of an unrestricted row. It is not sufficient. Full row rank of S permits the EM linear solve but does not ensure full rank of A. These are standard local arguments in an affine phase-retrieval setting, not a new identifiability theorem [1]. Sparse channel constraints require their own parameterization and Jacobian; a path-count condition is not derived here.

A. Rician Information

For known complex variance $v > 0$ and $a > 0$, the amplitude score is

$$h(z, a, v) = -2a/v + (2z/v)I_1(2za/v)/I_0(2za/v). \tag{3}$$

Define $w(a, v) = \mathbb{E}[h(Z, a, v)^2]$, with expectation under the Rician observation law. For independent pilots the real Fisher matrix is

$$J = A^T \operatorname{diag}\{w(a_p, v)\}_{p=1}^P A. \tag{4}$$

Each magnitude observation contributes an outer product of rank at most one, not a full-rank complex information block. For finite $v > 0$ and nonzero amplitudes these weights are positive. At a zero field the regular amplitude-coordinate derivation is not used; first-order information about the Cartesian mean vanishes. When the assumptions hold, the rank and conditioning of J give a local diagnostic. Neither a small bound nor a well-conditioned matrix at the true channel guarantees that an iterative estimator reaches the correct solution.

B. What a Reference Does and Does Not Fix

With $B = 0$, g and $e^{j\phi}g$ have identical amplitudes for every ϕ. The Jacobian has the phase direction jg in its null space for nonzero g. A nonzero known reference can remove that symmetry, but arbitrary reference and pilot patterns still require a rank check. For example, if $b_p = c^T s_p$ for some fixed c, then $g' = e^{j\phi}(g+c) - c$ produces the same amplitudes. This can occur with a full-row-rank pilot matrix and a nonzero reference. RSR 0 dB in our experiments means a nonzero reference equal in expected power to one user; it is not the absent-reference case.

There is a separate calibration ambiguity. If the estimator assumes $B' = e^{j\phi}B$, then $G' = e^{j\phi}G$ gives

$$|G'S + B'| = |e^{j\phi}(GS + B)| = |GS + B|. \tag{5}$$

The likelihood therefore cannot identify a constant reference-phase error from these observations alone. With error measured in the original reference frame, even this otherwise exact rotated estimate has NMSE $|e^{j\phi} - 1|^2 = 4\sin^2(\phi/2)$. At 10° this is about 0.0304 (-15.2 dB). This is an illustrative exact-fit ambiguity, not a lower bound on every biased estimator. A smaller withheld loss cannot certify correct absolute phase or recover an unknown reference phase. Time-varying phase errors generally cannot be absorbed into one constant channel rotation. A separate static calibration diagnostic estimates a

shared positive amplitude multiplier and noise variance under stated assumptions. It cannot remove this unknown-phase ambiguity.

V. Filter Properties and Computation

Let $H = U\,\mathrm{diag}(\sigma_i)V^H \in \mathbb{C}^{m\times q}$, $m \geq q$, and

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \qquad \tau > 0. \quad (6)$$

Its singular values are $f_\tau(\sigma) = \sigma^3/(\sigma^2+\tau^2)$. Thus $0 \leq f_\tau(\sigma) \leq \sigma$ and

$$\|F_\tau(H)\|_F \leq \|H\|_F, \qquad \|F_\tau(H) - H\|_F \leq \sqrt{q}\,\tau/2. \quad (7)$$

The second inequality follows because $\sigma - f_\tau(\sigma) = \sigma\tau^2/(\sigma^2+\tau^2) \leq \tau/2$.

For a channel vector, let d_i count its occurrences in the Hankel matrix and define $\|g\|_D^2 = \sum_i d_i|g_i|^2 = \|\mathcal{H}(g)\|_F^2$. Antidiagonal averaging followed by lifting is the orthogonal projection onto Hankel matrices in Frobenius norm. Therefore for fixed realized $\eta \in [0,1]$ and $\tau > 0$,

$$g^+ = (1-\eta)g + \eta\mathcal{H}^\dagger F_\tau(\mathcal{H}(g)), \quad (8)$$

$$\|g^+\|_D \leq \|g\|_D, \qquad \|g^+ - g\|_D \leq \eta\sqrt{q}\,\tau/2. \quad (9)$$

This bounds the size of one correction, not its error relative to the unknown true channel. It does not prove a contraction between two different inputs, particularly when the controller changes η and τ with the input. It also does not prove convergence of the full estimator. For fixed H, $\tau \to 0$ gives $F_\tau(H) \to H$ and $\tau \to \infty$ gives zero. No low-noise limit is asserted because the learned threshold need not vanish with noise.

The matrix-solve shrinker is smooth for $\tau > 0$, including repeated singular values. However, the controller also uses ordered largest and tail singular values. These features need not be differentiable at every singular-value tie. Finite-gradient numerical checks do not establish global differentiability of the whole network.

A. Why the Rician Measurement Update Is an EM Step

Let the unobserved complex field be $Y = GS + B + W$ and write each entry as $Y_{np} = Z_{np}e^{j\Phi_{np}}$. The receiver observes Z, not the phase Φ. For a current predicted field M, the conditional phase density is

$$p(\phi \mid z, M) = \frac{\exp\{\kappa\cos(\phi - \arg M)\}}{2\pi I_0(\kappa)}, \qquad \kappa = 2z|M|/v. \quad (10)$$

This is a von Mises distribution. Its first circular moment gives

$$Q = \mathbb{E}[Y \mid Z, M] = Z\frac{I_1(\kappa)}{I_0(\kappa)}e^{j\arg M}. \quad (11)$$

At $M = 0$, the phase is uniform and $Q = 0$. These are conditional complex fields, not estimates of the mean magnitude. Since $|Y|^2 = Z^2$ when Z is fixed, expanding $\mathbb{E}[\|Y - GS - B\|_F^2 \mid Z, M]$ leaves $\|GS + B - Q\|_F^2$ plus terms independent of the candidate G. Least squares therefore gives $(Q-B)S^H(SS^H)^{-1}$. This derivation uses independent proper complex Gaussian noise and a supplied positive variance; it does not use a strong-reference or high-SNR approximation.

TABLE XIII
Leading work per stage, excluding initialization

Operation	Leading order
Predicted fields and EM products	$O(NKP)$
EM solve, current implementation	$O(NK^2 + K^3)$
Spectral controller features	$O(Kmq^2)$
Smooth matrix filter	$O(Kmq^2 + Kq^3)$
Antidiagonal averaging	$O(Kmq)$
Small controller	$O(K \cdot 178)$

B. Relaxed EM Is Not the Same as the Full Stage

For fixed posterior field Q and supplied variance $v > 0$, the negative EM surrogate, up to constants, is $\Psi(G) = \|GS + B - Q\|_F^2/v$. Its minimizer is G_{EM} when S has full row rank. The normal equations give

$$\Psi(G) - \Psi(G_{\mathrm{EM}}) = \|(G - G_{\mathrm{EM}})S\|_F^2/v. \quad (12)$$

For $G_\gamma = G + \gamma(G_{\mathrm{EM}} - G)$, the surrogate decrease is

$$\Psi(G) - \Psi(G_\gamma) = \gamma(2-\gamma)\|(G - G_{\mathrm{EM}})S\|_F^2/v \geq 0 \quad (13)$$

for $0 \leq \gamma \leq 2$. Thus even an over-relaxed step within this interval is a generalized EM step for this quadratic M-step, in exact arithmetic. The supplied-model likelihood is nondecreasing. This result does not extend automatically to the subsequent Hankel correction, nor does it guarantee channel-error reduction, global recovery or convergence of the complete learned estimator. The secondary GD core is not covered by this particular argument.

C. Operation Counts

For $S \in \mathbb{C}^{K\times P}$ and $G \in \mathbb{C}^{N\times K}$, forming SS^H costs $O(K^2P)$. Spectral initialization forms N weighted $(K+1)$-dimensional Gram matrices and eigendecompositions, costing $O(NP(K+1)^2 + N(K+1)^3)$. These are per observation case, not per stage. The inverse-trace noise proxy adds $O(K^3)$ once.

Here $q = \lceil N/2 \rceil$, $m = N - q + 1$. Caching the pilot-Gram factorization could remove repeated K^3 factorizations, but the quoted timings use the existing implementation. The filter avoids differentiating hard rank truncation; it does not remove the SVD needed for controller features. At large N, both dense spectral features and the dense solve scale cubically. A small parameter count is not a small operation count.

For HS-GS, with $R = q$ candidate ranks, $T_s = 25$ selection updates, $T_f = 100$ refit updates and $C = 4$ sweeps, the dominant repeated work is of order

$$(RT_s + T_f)\,[NKP + NK^2 + K^3 + CKmq^2]. \quad (14)$$

Selection uses fewer pilot columns and has additional scoring and initialization work. This expression is a work estimate, not an exact FLOP count or a prediction of the measured speedup. Bessel functions, library choices, memory traffic and batching also matter. At $N = 32$, HS-GS uses 400 selection and 100 refit updates, compared with ten learned stages. Accuracy at those different budgets, not equal iteration count, is the reported comparison.

TABLE XIV
ALL ORIGINAL MAIN-STUDY COMPARATORS, UNCHANGED

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
EM constant-output control	-9.19	5.61	11.42
GD adaptive (secondary)	-14.59	8.00	0.85
GD learned steps only	-10.44	2.73	21.15
GD global filtering	-12.57	5.38	18.37
GD constant-output control	-9.95	5.36	12.36
HS-GS, rank search	-12.74	525.62	14.25
EM, 10 updates	-10.23	4.10	20.64
EM, 100 updates	-10.28	17.01	23.57
Rician GD, 10 updates	-9.31	4.17	17.86
Rician GD, 100 updates	-10.29	17.40	23.50
Rician GD, stopping rule	-10.27	45.37	23.75
Amplitude GD, 10 updates	-8.80	3.29	20.21
Amplitude GD, 100 updates	-9.65	8.87	26.96
Amplitude GD, stopping rule	-9.61	20.93	27.11
Rician GD, hard Hankel	-11.06	47.39	20.93
Linearized least squares	-6.92	1.06	19.86

Accuracy averages seven primary SNR cells. Time is the original batch-one median at SNR 5 dB, RSR 12 dB. These are not the new OptShrink comparison's data or timing sample.

TABLE XV
ORIGINAL TRANSFER RESULTS: GAIN OVER HS-GS IN DB

Changed quantity	Gain	Pointwise 95% interval
$N = 8$	-0.106	$[-0.286, 0.077]$
$N = 16$	1.223	$[1.086, 1.361]$
$N = 64$	0.165	$[0.024, 0.307]$
$K = 2$	0.078	$[-0.073, 0.228]$
$K = 4$	1.529	$[1.415, 1.639]$
$P = 10$	2.334	$[2.136, 2.531]$
$P = 20$	1.427	$[1.273, 1.577]$
Noise input -3 dB	1.226	$[1.095, 1.359]$
Noise input $+3$ dB	1.050	$[0.901, 1.193]$

VI. EXISTING DIMENSION TESTS AND CONTROLLER INPUTS

The original study already contains one-factor-at-a-time tests, with 400 cases per condition and weights fixed. Except for the changed quantity, the reference condition is $N = 32, K = 3, P = 30$, SNR 5 dB and RSR 12 dB. The pilot sweep's smallest value is ten for three users, not six. It does not test the boundary $P = 2K$. The alternative channels use four clusters of ten rays within $\pm 5^\circ$, ten equal-power paths, or ten paths with geometrically decreasing powers. They match base-channel energy and are tested only at SNR 5 dB and RSR 12 dB. These alternatives are simulation populations, not measured scattering distributions.

The full controller receives a spectrum summary, residual/noise proxies and size/stage information. A large spectral tail can reflect either noise or real paths; residuals can reflect noise, a wrong channel, or a wrong receiver model. The features provide context but no unambiguous failure detector. No sign constraint makes the threshold monotone in SNR, RSR or path count. The threshold and mixing can compensate for each other, so we inspect the actual relative channel correction as well as both outputs.

TABLE XVI
ORIGINAL FOLLOW-UP AVERAGE GAINS IN DB, UNCHANGED

Comparison	Set 1	Set 2
New-data adaptive vs. global	2.252	2.185
New-data adaptive vs. no filtering	4.250	4.191
New-data adaptive vs. HS-GS	1.939	1.837
Original adaptive vs. supervised	1.015	1.012
Original adaptive vs. final-only	0.697	0.694
Fixed-open mix vs. original adaptive	0.064	0.066

A. Original Follow-up Comparisons

The earlier follow-up trains 15 models and evaluates two fresh sets, each with seven SNR settings and two difficult settings of 200 cases. The six primary comparisons adjust for multiple testing within each set. The two new-data adaptive runs use independently generated training data, one fixed training seed per dataset. They are distinct from the three main seeds sharing one dataset and from the new reviewer diagnostics.

The supervised recipe holds architecture, input pilots and training budget fixed but trains on channel targets with mean linear NMSE. Selecting its checkpoint by withheld likelihood reduces the gap to approximately 0.73 dB; pooling linear errors across SNRs reduces it to about 0.36 dB. It wins in some cases, including two paths. Its transfer from 21 to 30 pilots was less consistent in validation replay. This comparison does not establish a general benefit of withholding labels. The final-only comparison retrains that alternative and its separate timing sample gives 4.65 versus 2.05 ms; those values do not replace the main timing table. Fixing mixing at one retains all 188 stored parameters and worsens two of three seeds on two-path cases, despite its small average gain.

VII. ADDITIONAL REVIEWER DIAGNOSTICS

These checks were specified after the main study and use frozen selected EM models (seeds 9101–9103). No learned weights or original results were changed. Seven SNR cells and two stress cells each contain 32 fresh worlds: 288 distinct worlds. Eight input-mismatch conditions reuse the same 32 SNR 5 dB worlds, giving 544 condition-cases, not 544 independent worlds. Classical and learned methods share the channel, pilots and noise. All tests use $N = 32, K = 3, P = 30$; the main RSR is 12 dB. The extra cells have RSR 0 dB or two paths, at SNR 5 dB.

The small sample, post-review design and fixed checkpoints make these exploratory checks. The intervals resample whole cases with common weights across methods and learned seeds, separately within each cell. They are unadjusted, conditional intervals, not a new confirmatory success test. The original 400-case cells and their intervals remain unchanged.

A. Same Candidates, Different Rank-Selection Loss

HS-GS fits ranks 1–16 on the first 21 pilots using 25 EM updates and four Cadzow sweeps. Exactly the same candidate

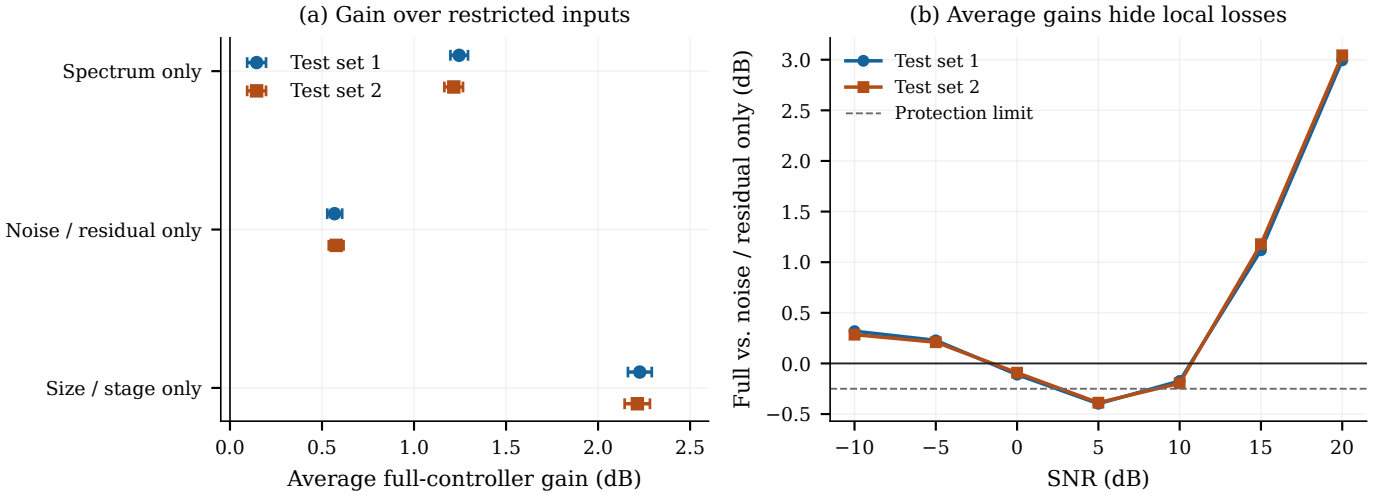


Fig. 5. Original smaller-budget input ablation, unchanged. Full inputs help the average but fail the declared 0.25 dB protection condition. These are separate models trained with 4,096 cases, 12 epochs and three seeds; they are not replacements for the main 16,384-case models.

TABLE XVII
EXPLORATORY SEVEN-SNR AVERAGES, 32 FRESH CASES PER CELL

Method	NMSE (dB)
EM100, no final filter	-10.162
EM100 + final hard rank six	-11.811
EM100 + final cubic filter	-11.648
EM100 + final OptShrink rank seven	-13.691
HS-GS, residual rank score	-12.778
HS-GS, Rician rank score	-12.961
Frozen learned model, stage-five prefix	-13.353
Frozen learned model, all ten stages	-14.243

channels are scored on the last nine pilots using either squared magnitude residuals or Rician negative log-likelihood. Each selected rank is refitted with 100 updates and all 30 pilots. This changes only the selection score, not the fit budget or observation count. A replay against the original residual-ranked solver agreed exactly on a fresh case; an independent likelihood check agrees with SciPy's Rician density up to observation-only constants.

Across the seven SNR cells, Rician scoring improves HS-GS by 0.182 dB, with interval $[0.104, 0.267]$ dB. The frozen learned estimator retains a gain of 1.282 dB, with interval $[1.092, 1.485]$ dB. This score difference explains part, but not all, of the gap in this small check.

B. A Stronger Final-Filter Control

A separate bank of 72 development worlds (eight per SNR/stress cell) selects one global setting for each final-filter method. Selection uses only the withheld Rician likelihood after EM100 on 21 pilots. Hard truncation and OptShrink search ranks 1–15; cubic filtering searches relative thresholds $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$. Every method also has a no-correction option. Candidate counts differ and the grids are small; this is not an exhaustive equal-search-budget comparison. The selected settings are rank six, relative threshold 0.3 and OptShrink rank seven, respectively. The settings are frozen before reading test observations, then used after EM100 with all 30 pilots. None use true channels for tuning.

OptShrink uses the empirical spectral-tail transform in the original algorithm [2], including rectangular zero directions and reduced tail dimensions. A rank equal to full capacity has no tail and is excluded; no correction is a separate candidate. At a tied boundary within a relative squared gap of 10^{-13}, the affected weight is set to its zero-weight limit. Primitive checks cover zero tail, scaling, transpose symmetry, direct rectangular trace evaluation, zero input and the tie rule. These are numerical safeguards, not a new optimality result.

In the matched one-pass test, OptShrink beats the cubic filter by 2.042 dB on the seven-SNR average. Cubic shrinkage is therefore not generally the best structural operation. The complete frozen learned estimator has a 0.552 dB average gain over EM100 plus OptShrink, with conditional interval $[0.432, 0.673]$ dB. It loses 1.040 dB at SNR 20 dB and 2.408 dB at RSR 0 dB, but gains 1.830 dB on two-path cases against this globally selected rank-seven control. That last result does not remove the two-path loss against per-case rank-selected HS-GS.

These comparisons have different update budgets, training/development budgets and filtering schedules. They do not identify the best possible OptShrink-based estimator or reproduce iterative LRHD. No controlled runtime comparison was added; the original 64-fold timing ratio applies only to the original HS-GS implementation.

C. Reference and Noise Inputs

The physical observations stay fixed. For amplitude errors the estimator receives $B_{\text{in}} = aB_{\text{true}}$ with $a = 0.9$ or 1.1; these are amplitude, not power, multipliers. Constant phase errors use $B_{\text{in}} = e^{j\delta}B_{\text{true}}$ with $\delta = \pm 10°$. The ramp uses $B_{\text{in},np} = B_{\text{true},np}e^{j\delta p/(P-1)}$, $p = 0, \ldots, P-1$: the assumed phase error grows from zero to $\pm 10°$. Noise errors supply $v_{\text{in}} = v_{\text{true}}10^{\Delta/10}$, $\Delta = \pm 6$ dB. These are input-mismatch stress tests, not measured hardware drift. They test errors

TABLE XVIII
INPUT MISMATCH AT SNR 5 DB: LOWER NMSE IS BETTER

Assumed input error	Learned (dB)	HS-Rician (dB)	Learned loss vs. matched (dB)
None	-14.043	-12.496	0.000
B amplitude -10%	-11.613	-9.799	2.430
B amplitude $+10\%$	-12.252	-11.328	1.791
B phase -10°	-11.685	-10.730	2.358
B phase $+10^\circ$	-11.572	-10.564	2.471
Phase ramp $0 \to -10^\circ$	-13.267	-12.029	0.776
Phase ramp $0 \to +10^\circ$	-13.204	-11.774	0.839
Noise variance -6 dB	-13.806	-12.567	0.238
Noise variance $+6$ dB	-10.995	-12.404	3.048

relative to a fixed true reference, not a dynamically changing atomic receiver or a joint calibration algorithm.

Supplying noise variance 6 dB too high worsens the learned estimate by 3.048 dB relative to matched inputs, and makes it 1.409 dB worse than Rician-ranked HS-GS on these paired cases. This is a new observed failure, not evidence of general variance robustness. Amplitude errors also cause clear losses. Constant reference-phase errors rotate every estimated channel by the same phase; replay verifies the equivariance and unchanged phase-aligned errors. Alignment uses truth only as a diagnostic and is not an available inference correction. Absolute-frame errors in the table remain the reported errors.

D. Depth and Controller Trajectories

The stage-five output is taken from the unchanged ten-stage forward pass, with its original stage feature $(t+1)/10$. It is not a five-stage model retrained with $(t+1)/5$. Completing ten stages improves the seven-SNR average by 0.890 dB, with interval $[0.815, 0.964]$ dB; it does not help every cell. This prefix experiment does not determine an optimal depth or answer whether a trained twenty-stage model is better.

At the ordinary SNR 5 dB condition the mean relative threshold falls from about 0.915 at stage one to 0.362 at stage ten. Mean mixing is about 0.99996. Thus the useful adaptation is primarily in threshold strength, not abstention. Large relative corrections in a failure condition do not prove which feature or prior caused the failure.

VIII. CONTROLLED IN-LOOP OPTSHRINK COMPARISON

This experiment uses the frozen original three EM models. A separate development set has 32 worlds at each of seven SNRs. For both in-loop and final-only OptShrink, the candidate grid includes ranks 1–15, mixing 0.25, 0.5 or 1, update budgets 10, 20, 50 or 100, and no filtering. Candidates fit 21 pilots; the other nine score Rician likelihood. Noise-dependent settings are frozen before generating 128 fresh worlds at each SNR and at each of two stress settings, giving 1,152 test worlds. All test methods use all 30 pilots. Reference strength is 12 dB and channels have three to seven paths, except in the labelled stress cells.

Unlike the first final-filter diagnostic, this control shrinks the lifted channel after each EM update and averages its antidiagonals. It is an OptShrink-based Rician reconstruction, not full LRHD on directly observed noisy Hankel matrices. Hankel lifting and nonlinear reconstruction produce dependent errors, so matrix-denoising optimality guarantees are not claimed. Both noise-dependent and global settings are retained in the results.

The retained narrow-population figure includes these saved results and the added one-sweep comparison. The per-SNR intervals are conditional paired-case intervals, with no per-SNR multiplicity adjustment.

The selected in-loop control uses ten iterations up to SNR 10 dB and twenty at 15 and 20 dB. The learned model loses 0.143 dB on the declared seven-SNR average. It gains 0.723 dB at -10 dB but loses 0.796 dB at 20 dB. Its severe-error rate is 0.89%, versus 5.02% for the selected in-loop method. At -10 dB these rates are 6.25% and 35.16%; the other primary SNR cells have none in this sample. The lower mean per-case linear error of the learned method is a different metric, not a reason to replace the original primary metric after seeing the results.

At RSR 0 dB and SNR 5 dB, the selected in-loop control has NMSE -5.026 dB, versus -7.069 dB for learned and -8.307 dB for HS-Rician. However, its 50- and 100-iteration controls reach -8.790 and -8.981 dB. The apparent learned advantage over the selected control therefore does not establish superiority over the in-loop family at weak references. For two paths, HS-Rician reaches -17.833 dB, learned -16.814 dB and selected in-loop -16.407 dB. No algorithm wins every comparison.

End-to-end single-thread CPU timing at SNR 5 dB includes initialization, noise-level lookup and the selected reconstruction. Mean times are 5.67 ms for one learned model, averaged across three trained models, and 6.20 ms for selected in-loop OptShrink. Final-only filtering takes 3.99 ms and HS-Rician 356.65 ms in this timing sample. These are not ensemble inference times, embedded-device measurements or GPU benchmarks. The original 64-fold ratio applies only to the earlier HS-GS comparison. Saved outputs were recalculated; independent OptShrink primitive replays agree within 5×10^{-15} and selected learned replays agree exactly.

IX. STATIC NOISE AND REFERENCE ESTIMATION

This exploratory check estimates a single positive reference-amplitude multiplier a shared by the array and a shared complex noise variance v. Reference shape and phase remain supplied. It does not estimate per-antenna drift or choose new measurements. With posterior field Q, let $R = I - S^H(SS^H)^{-1}S$. One conditional least-squares update is

$$a^+ = \mathrm{clip}_{[0.25,4]} \frac{\mathrm{Re}\langle BR, QR\rangle_F}{\|BR\|_F^2}, \tag{15}$$

$$G^+ = (Q - a^+B)S^H(SS^H)^{-1}, \tag{16}$$

$$v^+ = \mathrm{clip}_{[10^{-6},10^3]}\, \mathrm{mean}\{Z^2 + |M^+|^2 - 2\,\mathrm{Re}[(M^+)^*Q]\}, \tag{17}$$

where $M^+ = G^+S + a^+B$. Bounds use the normalized simulation units and are algorithm assumptions, not measured receiver limits. If $BR = 0$, the gain is confounded with the

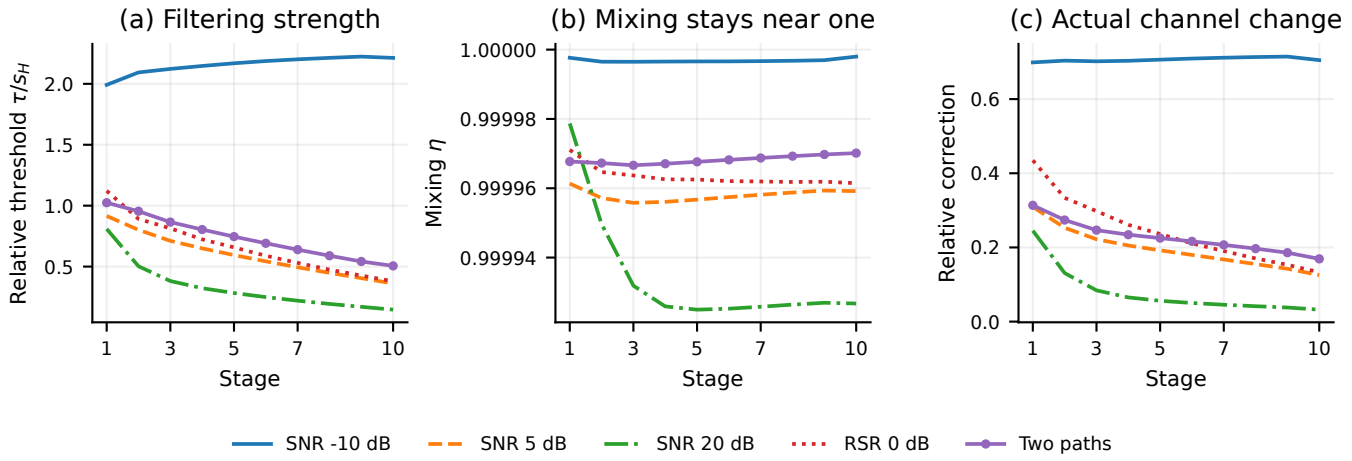


Fig. 6. New frozen-model trajectories, averaged over 32 cases, three model seeds and users. Main SNR curves use RSR 12 dB; weak-reference and two-path curves use SNR 5 dB. All other settings are unchanged. Mixing stays near one, while threshold and actual correction vary. Curves describe outputs; they do not establish causal feature importance or a failure detector.

channel and the joint fit is rejected. This denominator check is not a global identifiability proof for the full nonlinear problem.

Three choices—unchanged inputs, noise estimation, and joint gain/noise estimation—are fitted on 21 pilots for 5, 20 or 100 iterations. The other nine choose a setting, which is refitted on all 30 observations. The selection loss includes $\log v$:

$$\ell_{\text{nuisance}} = (z - |\mu|)^2/v - \log I_{0e}(2z|\mu|/v) + \log v. \quad (18)$$

Only $-\log(2z)$ may be omitted when comparing different variances. All nine candidates and the final refit count toward processing cost. The calibration preprocessing averages 45.05 ms per case in this CPU check, before the unchanged learned estimator runs. It is not a free robustness improvement. An EM control with the same held-out stopping choice separates stopping effects from nuisance estimation. Corrected inputs are also passed to the unchanged learned model; there is no retraining on these cases.

There are 32 worlds per SNR/reference cell, at SNR $-5, 5, 15$ dB and RSR 0 or 12 dB: 192 independent worlds. Seven assumed-input conditions reuse them, giving 1,344 condition-cases, not independent worlds. The conditions are correct inputs, variance errors of ± 6 dB, reference-amplitude errors of $\pm 10\%$, joint $+10\%/+6$ dB error, and constant phase error $+10^\circ$.

Average improvement does not make the correction safe in every cell. With weak reference at SNR -5 dB and a $+6$ dB variance error, it worsens learned NMSE by 1.048 dB. At SNR 15 dB with strong reference, a $+10\%$ amplitude error is largely corrected: NMSE changes from -16.346 to -24.107 dB; the true-input diagnostic is -24.253 dB. These are simulation input errors, not observed hardware behavior. Unknown constant phase remains unresolved because channel and reference can rotate together. Likelihood improvement cannot certify absolute phase or reduced channel error. The original known-calibration estimator remains the primary method; this check is not a universal calibration remedy.

TABLE XIX
STATIC CALIBRATION: GAIN OF CORRECTED OVER SUPPLIED INPUTS

Input condition	Mean gain (dB)	Worst cell gain (dB)
Correct inputs	-0.013	-0.054
Variance -6 dB	0.832	0.138
Variance $+6$ dB	0.997	-1.048
Amplitude -10%	1.863	-0.072
Amplitude $+10\%$	1.449	-0.413
Joint $+10\%/+6$ dB	2.872	-0.746
Phase $+10^\circ$	-0.008	-0.048

Equal average over six SNR/RSR cells. Positive gain means lower error. These are descriptive point estimates, not a simultaneous robustness guarantee.

X. Tight Pilot Counts and Weak References

The new pilot sweep has $P = 6, 8, 10, 12, 20, 30$, $K = 3$, SNR $-5, 5, 15$ dB, RSR 12 dB and 32 fresh worlds per cell, or 576 worlds. Learned weights and OptShrink settings are transferred from $P = 30$, without tuning on these tests. HS-Rician fits ranks on $\text{round}(0.7P)$ pilots, scores the remainder, then refits all P. Its rank-fitting subset can be below $2K$ even when P reaches it. Structural coupling across array elements means the unrestricted single-row count is not, by itself, an identifiability theorem for the structurally constrained model.

At SNR 5 dB, learned NMSE is -2.197 dB with six pilots and -14.332 dB with thirty. At SNR -5 dB and six pilots it is 4.089 dB, showing substantial failure. All sampled nonzero-reference full-data row Jacobians have rank six. Nevertheless, at SNR 5 dB the median smallest singular value is 0.172 with six pilots and 2.488 with thirty. Local full rank is not enough for stable finite-noise estimation or global recovery. The cells use different independent worlds; the method comparisons within each cell are paired.

A second check shares the same channel, pilots and complex noise across absent reference and RSR $-10, 0, 5, 12$ dB, at SNR 5, 20 and 60 dB. There are 32 independent worlds per SNR and 480 condition-cases. The SNR 5 dB reference

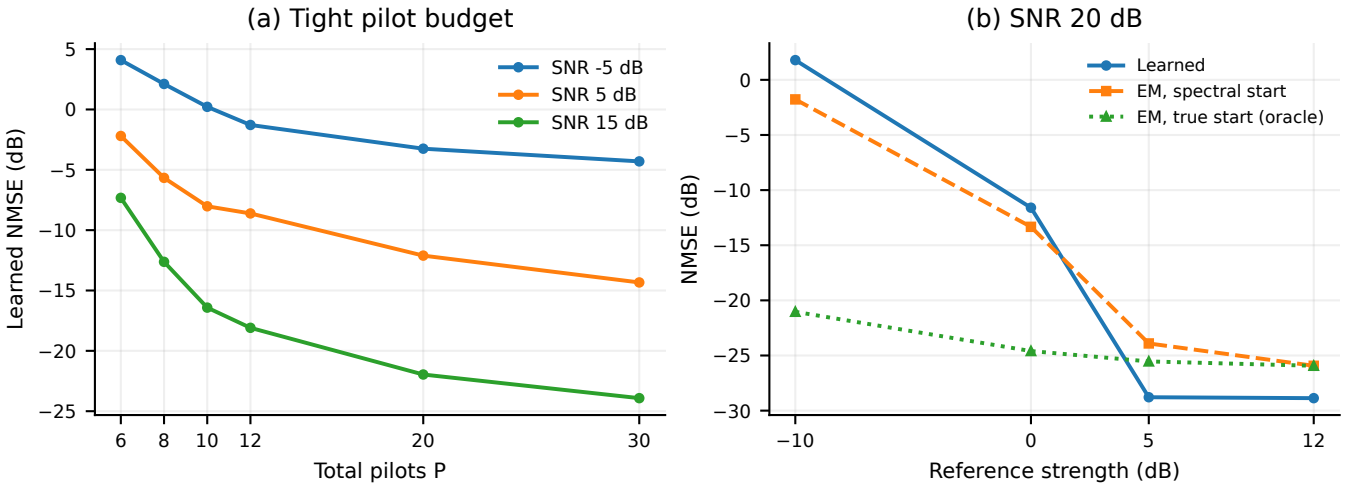


Fig. 7. Exploratory checks, 32 worlds per cell. Left: frozen learned model under changing pilot count. Right: EM initialization diagnostic with the same channel, pilots and noise across reference levels. The true-start curve uses unavailable ground truth and is not a deployable estimator.

check shares a generation seed and channels with one tight-pilot cell. The two diagnostic phases are not independent replications of each other and are not pooled for uncertainty estimates. At zero reference, sampled Jacobians have rank five and absolute phase is ambiguous. At nonzero reference they have rank six in this sample. At SNR 20 dB and RSR -10 dB, ordinary EM gives -1.773 dB NMSE from spectral initialization but -21.036 dB when initialized at truth. The latter is an oracle diagnostic only. The gap supports an initialization or optimization difficulty; it does not prove global identifiability or exclude controller problems.

A. A Nearly Noiseless Failure

SNR 60 dB lies far outside the training range. Even at RSR 12 dB the frozen learned model gives -6.647 dB, while EM100 gives -65.715 dB. A post-hoc intervention on the same 32 cases disables only learned filtering and retains the learned step sizes; its NMSE is -44.020 dB. Starting the full learned model at the true channel still gives -6.653 dB. Thus filtering can spoil an already accurate estimate in this setting; the failure is not explained solely by a bad initializer. Independent NumPy arithmetic for the no-filter intervention agrees with the PyTorch result within 5×10^{-15}. This is an out-of-range failure and a post-hoc diagnostic, not an independently confirmed switching policy. No low-noise consistency or generally safe filtering claim is made.

XI. Separately Trained Depth, Width and Training Controls

This reduced-budget study answers the depth and capacity questions using new training, not a prefix of the original network. Each dataset contains 4,096 training and 448 independent validation worlds. Training uses 12 epochs, batch size 32 and Adam with learning rate 0.001. There are seven SNRs from -10 to 20 dB, RSR 0, 5, 12 and 20 dB, and either random three-to-seven-path or fixed two-path channels. This gives 56 conditions, with $N = 32, K = 3, P = 30$. Unlike the original training mixture, this one includes RSR 0 dB. Its scores must not be pooled with the original study.

Seven variants share dataset zero and three initialization seeds: the ten-stage, width-16 baseline; depths five and twenty; widths eight and 32; two experts; and curriculum training. The baseline also trains on two additional independent datasets with two initialization seeds each, giving 25 jobs and 300 completed epochs. Initialization seeds do not create independent training populations. Training has no channel targets. Validation selects the checkpoint using the equal-condition average of withheld Rician likelihood, with a fixed 21/9 pilot split. The final epoch is also evaluated, not substituted when its test error is better.

Ordinary variants share example order and pilot masks at each epoch. Depth changes the number of learned steps and uses $(t + 1)/T$ as its stage input. The two-expert model combines two width-16 controller outputs with a learned softmax gate driven by the three spectral features. It has 374 parameters, compared with 364 for width 32. The experts start at relative thresholds 0.05 and 0.2 rather than the plain controller's 0.1. They do not recover distinct physical channel components. Architecture and initialization both change; any gain cannot be attributed to gating alone.

The curriculum presents RSR 12/20 examples twice per epoch for six epochs, then RSR 0/5 examples twice per epoch for six epochs. Total exposure per example matches ordinary training. The repeated visits share that epoch's pilot mask, so the number of distinct masks per example differs. This is not an isolated test of ordering alone. Neither channel truth nor test errors determine this schedule.

After all checkpoints were fixed, 64 fresh worlds per condition were generated, giving 3,584 paired test worlds. All 25 selected and 25 final models use the same observations, together with EM100 and the previously frozen in-loop OptShrink rule. That rule transfers from its earlier development mixture; it is not retuned on these new tests. A documented test-seed amendment was made before generation to avoid overlap with

diagnostic seeds. Training and selection were unchanged.

A. Accuracy and Cost

The reported macro-average gives each condition and initialization equal weight in dB. Five stages are faster, but lose 1.977 dB against ten. Twenty gain 0.181 dB while increasing mean time from 8.04 to 15.60 ms. Their severe-error rate is also higher: 13.46% versus 10.28% in this 56-condition mixture. The depth comparison is therefore not a simple "more is better" result. Twenty stages improve the RSR 0 dB average by 1.547 dB but worsen one low-SNR cell by 1.586 dB. The absolute severe-error rates use a broader, weaker-reference mixture than the original experiment and are not evidence of a change in its saved results.

Width eight loses 0.731 dB and has almost unchanged processing time: matrix operations, not controller size, dominate this implementation. Width 32 gains 0.065 dB on average and reduces severe errors, but loses 0.970 dB in its worst cell. The two-expert gain is only 0.023 dB overall and 0.179 dB for two-path cells; width 32 gains 0.286 dB on those cells. The experts therefore do not demonstrate a useful advantage over the similarly sized plain controller in this check.

The curriculum gains 0.206 dB overall and 0.255 dB at RSR 0 dB, but loses 2.353 dB at SNR 20 dB, RSR 5 dB and two paths. It does not solve weak-reference or few-path estimation generally. No tested variant improves every cell. The original primary estimator is retained; these test results are not used to choose a replacement.

Timing uses one CPU thread, double precision and batch size one, includes initialization, and repeats 32 shared SNR 5 dB/RSR 12 dB worlds three times in shuffled model order. Means average individual-model inference, not ensemble inference. This timing batch is separate from the original and OptShrink timing batches; cross-batch speed ratios are not claimed. The 25-job training invocation took about 103 minutes on the local CPU, including data preparation and validation, but excluding fresh testing.

B. Variability and Scope

The three independent training datasets give baseline two-initialization macro-averages of -12.450, -12.124 and -12.229 dB. Their individual initialization scores are $(-12.391, -12.509)$, $(-11.824, -12.425)$ and $(-12.143, -12.315)$ dB. These few replicates expose variability but do not estimate broad population uncertainty reliably. All use the same test worlds; repeated predictions are not independent channel realizations.

The paired intervals resample 64 whole cases within each condition 2,000 times, retaining paired methods and averaging the three fixed-model scores. They are exploratory, conditional and unadjusted across six comparisons. Small intervals do not remove uncertainty from training or prove an architecture optimum. Most selected checkpoints are the final epoch; 12 epochs do not establish convergence.

The transferred OptShrink rule gives -12.150 dB and 12.11% severe errors; EM100 gives -8.784 dB and 25.53%. The new baseline's -12.390 dB score cannot reverse the earlier controlled OptShrink result: both the training mixture and test weighting differ, and the classical settings were not developed for this mixture. These are transfer controls, not a new claim of tuned-classical superiority.

C. Saved-Result Audit

All 25 histories, 300 epochs, 50 checkpoint hashes and 2,912 prediction files were checked. The latter comprise 1,400 selected-model, 1,400 final-model and 112 classical files. Errors, severe-error rates and headline averages were recalculated from predictions and truth. Matching orders and masks were checked with the stated curriculum exception. One world per test condition was regenerated from its seed; selected and final models were replayed individually on a saved case. The largest replay difference was below 1.2×10^{-15}. Seventeen focused tests cover the new numerical/training/seed checks and OptShrink implementation. The release retains these audit tools and saved observations, not just summary scores. This checks the reported computation; it is not external replication or real-receiver validation.

XII. Related Structural and Sparse Methods

OptShrink estimates singular-value weights from the observed noise spectrum [2]. LRHD combines that principle with repeated Hankel projection [3]; neither our first one-pass diagnostic nor the new EM/OptShrink adaptation reproduces full LRHD. In both cases the starting problem is noisy matrix denoising, not recovery from affine magnitudes. The noise on a Hankel matrix formed from an iterated channel estimate need not meet asymptotic independent-entry assumptions. Neither those guarantees nor universal optimality are asserted here.

Structured alternating least squares reconstructs low-rank matrices from linear measurements, including Hankel structure [4]. HSGD addresses partially observed Hankel matrices with sparse gross corruptions [5]. Their losses, observations and guarantees cannot be transferred unchanged to our nonlinear Rician model. They remain important alternative constructions, not irrelevant methods.

Noncoherent compressive channel estimation combines phase retrieval and sparse reconstruction using a designed sensing cascade [6]. GP-bandit beam training uses adaptive near-field beams, Rician-debiased pseudo-amplitudes and sparse phase retrieval [7]; it has no known additive reference. Partially coherent matching pursuit retains complex observations with a shared unknown phase within each packet [8]. That information is absent from our magnitude-only inputs. A fair benchmark must adapt these methods to the same observation and pilot budget rather than importing their published scores.

The multi-shot Rydberg study [9] concerns signal-presence detection with Rician shots. It motivates accounting for observation statistics and shot budgets, but supplies neither channel-estimation results nor a heavy-tailed noise model for this experiment.

TABLE XX
SEPARATELY TRAINED CONTROLS: ALL 56 CONDITIONS, EQUAL-WEIGHT DB AVERAGE

Variant	Parameters	Selected NMSE (dB)	Final NMSE (dB)	Gain (dB)	Worst-cell gain (dB)	Severe (%)	Time (ms)
Ten stages / width 16	188	-12.390	-12.297	0.000	0.000	10.28	8.04
Five stages	183	-10.413	-10.413	-1.977	-9.540	6.48	4.25
Twenty stages	198	-12.571	-12.571	0.181	-1.586	13.46	15.60
Width 8	100	-11.658	-11.658	-0.731	-3.070	14.14	8.02
Width 32	364	-12.454	-12.454	0.065	-0.970	4.91	8.04
Two experts	374	-12.413	-12.385	0.023	-0.931	8.31	8.55
Curriculum	188	-12.596	-12.634	0.206	-2.353	6.84	8.03

All rows use dataset zero, three initialization seeds, 4,096 training cases and 12 epochs. Gain is relative to ten stages/width 16; positive favors the variant. Severe rates and gains use selected checkpoints. Time is individual-model mean inference time, not an ensemble time. This protocol differs from the original study.

TABLE XXI
INITIALIZATION VARIABILITY AND CONDITIONAL GAINS IN THE REDUCED-BUDGET STUDY

Variant	Selected seed scores (dB)	Selected epochs	Gain interval (dB)	RSR 0 gain	Two-path gain
Ten stages / width 16	-12.391, -12.509, -12.269	12, 12, 11	—	0.000	0.000
Five stages	-10.779, -10.906, -9.554	12, 12, 12	$[-2.137, -1.849]$	-1.500	-1.943
Twenty stages	-12.382, -12.579, -12.751	12, 12, 12	$[0.062, 0.401]$	1.547	0.075
Width 8	-11.906, -11.288, -11.781	12, 12, 12	$[-0.779, -0.710]$	-0.158	-0.696
Width 32	-12.546, -12.659, -12.158	12, 12, 12	$[0.050, 0.087]$	0.115	0.286
Two experts	-12.532, -12.478, -12.228	11, 12, 12	$[0.014, 0.035]$	0.100	0.179
Curriculum	-12.676, -12.903, -12.209	11, 12, 12	$[0.153, 0.292]$	0.255	0.117

Initialization seeds are 9201, 9202 and 9203 in order. Intervals are unadjusted conditional 95% paired-case intervals, not intervals over training datasets or initialization seeds. Subgroup gains average all matching conditions; they do not imply improvement in every matching cell.

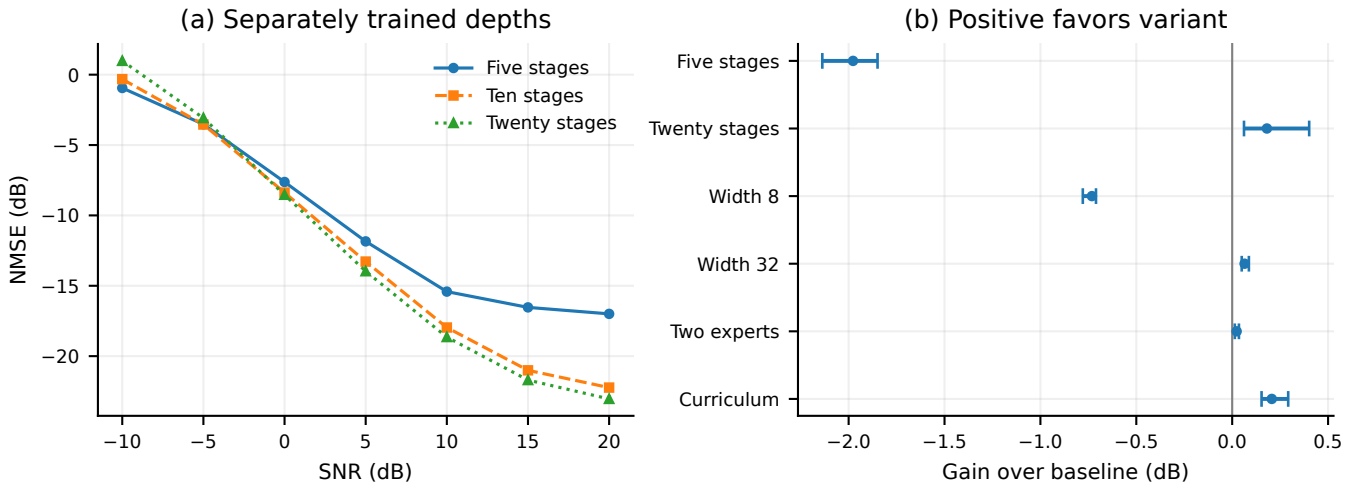


Fig. 8. New reduced-budget training study. Left: selected-model NMSE, averaging four reference levels, two path settings and three initializations at each SNR. Right: gain over ten stages/width 16 across all 56 conditions. Bars are unadjusted conditional 95% paired-case intervals; they exclude training-data and initialization uncertainty.

XIII. WHAT IS STILL NOT TESTED

Depth. The new 5/10/20-stage comparison uses separately trained models, unlike the original prefix diagnostic. It has a fixed small training budget, not an exhaustive search or a convergence test. An optimal depth for larger datasets or other channel populations is not known.

Noise and hardware. The original model has independent complex Gaussian noise before magnitude extraction. Incorrect variance inputs do not test non-Gaussian noise. Finite shot count alone does not imply heavy-tailed noise, and no cited detection model establishes that assumption for this receiver. Optical readout, atomic dynamics and hardware data remain untested. A Rician loss is not automatically matched to an alternative optical or multi-shot measurement statistic.

Structure. Cubic shrinkage is one design choice, not a proved optimal denoiser. The new in-loop comparison covers one OptShrink-based reconstruction. Full LRHD, structured ALS and robust Hankel completion would require their own adapted measurement models and independently selected settings. Existing hard, global, adaptive and no-filtering controls

do not exhaust these choices. An angle-domain sparse phase-retrieval method is also a relevant missing comparison, particularly for two-path channels. No superiority over those methods, or over published URformer, is claimed.

Training. Two-path and RSR 5 dB cases already occur in the original training; RSR 0 dB does not. The new smaller study includes RSR 0 dB, and its curriculum has mixed results. No safety-critical protection guarantee follows from average gains. The original pointwise intervals condition on fixed models and shared training data; the fresh-data follow-up is limited and does not estimate broad training-population uncertainty.

Array geometry. A two-dimensional array may suggest multilevel or block-Hankel structure, but its lifting dimensions, normalization, computational cost and training transfer must be defined anew. No 2D array result is reported.

XIV. Initialization and Numerical Record

For array element n, form $C_n \in \mathbb{C}^{P\times(K+1)}$ with rows $[s_p^T, b_{np}]$. Let u_n be the dominant eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$ and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{19}$$

The first K components of $c_n u_n \exp[-j \arg(u_{n,K+1})]$ initialize the channel row. Only the 21 input pilots enter this calculation during training. The withheld columns do not enter the initial estimate.

The original 43 focused tests cover likelihood and derivative checks, noise scaling, finite gradients, repeated singular values, pilot-split isolation, single/double precision agreement and deterministic recovery. The submission audit recalculated 1,120 main, 702 follow-up and 432 input-study result files. Three primary checkpoints replayed eight saved observations exactly; 24 input-study checkpoint replays were also checked. This is not a replay of every historical experiment or an independent hardware validation.

One finite-input SVD failure occurred in the original classical HS-GS test at $N = 64$. The same matrix was recomputed with SciPy's alternative SVD driver; relative reconstruction error was below 3×10^{-15}. The trial was retained with unchanged inputs and iteration budget. Original software versions were Python 3.12, NumPy 2.4.6, SciPy 1.17.1 and PyTorch 2.13.0. The expanded release includes saved inputs, truth for evaluation, predictions, selected/final checkpoints, training histories, settings, source hashes and figure scripts for the reported studies. The separate Overleaf archive contains manuscript files, not experimental data. No GPU agreement or hardware measurement is claimed.

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